STAT/MA 41600
In-Class Problem Set #28: October 23, 2026
Solutions by Mark Daniel Ward

Problem Set 28 Answers

1a. We compute $E(X) = \int_0^8 \int_0^{5-5x/8} (x)(1/20)\, dy\, dx = 8/3$, or alternatively, $E(X) = \int_0^5 \int_0^{8-8y/5} (x)(1/20)\, dx\, dy = 8/3$, or alternatively, $E(X) = \int_0^8 (x)(1/32)(8-x)\, dx = 8/3$.
1b. We compute $E(Y) = \int_0^8 \int_0^{5-5x/8} (y)(1/20)\, dy\, dx = 5/3$, or alternatively, $E(Y) = \int_0^5 \int_0^{8-8y/5} (y)(1/20)\, dx\, dy = 5/3$, or alternatively, $E(Y) = \int_0^5 (y)(2/25)(5-y)\, dy = 5/3$.

2a. We compute $E(X) = \int_0^\infty \int_0^\infty (x)(14e^{-2x-7y})\, dy\, dx = 1/2$, or alternatively, $E(X) = \int_0^\infty (x)(2e^{-2x})\, dx = 1/2$.
2b. We compute $E(Y) = \int_0^\infty \int_0^\infty (y)(14e^{-2x-7y})\, dy\, dx = 1/7$, or alternatively, $E(Y) = \int_0^\infty (y)(7e^{-7y})\, dy = 1/7$.

3. We compute $E(X) = \int_0^5 \int_0^5 (x)(4/1875)(25-xy)\, dy\, dx = 20/9$, or alternatively, $E(X) = \int_0^5 (x)(4/15 - 2x/75)\, dx = 20/9$.

4a. We compute $E(X) = \int_0^5 \int_0^2 (x)(1/25)(5-x)(2-y)\, dy\, dx = 5/3$, or alternatively, $E(X) = \int_0^5 (x)(2/25)(5-x)\, dx = 5/3$.
4b. We compute $E(Y) = \int_0^5 \int_0^2 (y)(1/25)(5-x)(2-y)\, dy\, dx = 2/3$, or alternatively, $E(Y) = \int_0^2 (y)(1/2)(2-y)\, dy = 2/3$.

5a. We compute $E(X) = \int_0^\infty \int_x^\infty (x)(63e^{-2x-7y})\, dy\, dx = 1/9$, or alternatively, $E(X) = \int_0^\infty (x)(9e^{-9x})\, dx = 1/9$.
5b. We compute $E(Y) = \int_0^\infty \int_x^\infty (y)(63e^{-2x-7y})\, dy\, dx = 16/63$, or alternatively, $E(Y) = \int_0^\infty (y)(63/2)(e^{-7y} - e^{-9y})\, dy = 16/63$.